

# Arnab Maity

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github.com/arnab2113

## PROFILE

B.Tech Computer Science & Engineering student at Sister Nivedita University with a passion for Full Stack Development and Generative AI. Skilled in building scalable web applications and AI-powered solutions using React.js, Next.js, Node.js, Express.js, MongoDB, PostgreSQL, Docker, and AWS. Eager to learn, solve real-world problems, and contribute to innovative software projects.

## EDUCATION

| Degree           | Institute                      | Year      | CGPA/% |
|------------------|--------------------------------|-----------|--------|
| B.Tech CSE       | Sister Nivedita University     | 2022–2026 | 7.6    |
| Higher Secondary | Gopalnagar Biharilal Vidyapith | 2022      | 91%    |
| Secondary        | Gopalnagar Biharilal Vidyapith | 2020      | 70%    |

## TECHNICAL SKILLS

- **Frontend:** React.js, Next.js, HTML, CSS, JavaScript
- **Backend:** Node.js, Express.js
- **Databases:** MongoDB, PostgreSQL
- **Cloud:** Docker, AWS
- **Programming:** C++, JavaScript
- **Other:** Git, GitHub, Graph Theory, Algorithms, OpenMP, Generative AI

## EXPERIENCE & FREELANCE

Oasis Infobyte (AICTE OIB-SIP Internship) May 2025 – Jun 2025

- Developed responsive and interactive web applications using HTML, CSS, JavaScript, and React.js.
- Built reusable UI components with clean, responsive layouts.
- Implemented modern frontend practices to improve user experience and application performance.
- Used Git and GitHub for version control and collaboration.
- Improved debugging and problem-solving skills through hands-on development.

Freelance Frontend Engineer Freelance

- Built responsive React.js user interfaces from client requirements.
- Delivered pixel-perfect, performance-focused frontend implementations.
- Maintained clean code and cross-device compatibility.

## RESEARCH PROJECT

Pebble Traversal-Based Fault Detection for Digital Microfluidic Biochips

Technologies: C++, OpenMP, Graph Theory, Algorithms

- Contributed to implementation and evaluation of Pebble Traversal-based fault detection for DMFBs.
- Optimized droplet traversal paths using Graph Theory.
- Implemented Manhattan distance-based rerouting to bypass faulty electrodes.
- Compared traversal time and fault detection efficiency across biochip configurations.
- Gained hands-on experience in algorithm optimization and research-oriented software development.